

2024



All-In Migration: P to the Cloud Using

Phillips 66 migrated its file storage and agility and reducing costs.



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330,000 files

migrated in 1 hour

300+ TB

of data migrated in less than 1 month

80%

reduction in storage costs

132 TB

migrated in 1 week to exit data center

Overview

With global operations and more than 14,000 employees worldwide, [Phillips 66](#) must manage a complex and vast data infrastructure and continually seeks ways to optimize it. As part of its cloud migration journey, the energy company wanted to migrate its on-premises NetApp ONTAP environment—which hosted its enterprise applications and corporate file shares—to Amazon Web Services (AWS). It aimed to achieve a familiar on-premises experience, but with the agility, scalability, and security of AWS.

To bring its production and disaster recovery (DR) workloads to the cloud, Phillips 66 adopted [Amazon FSx for NetApp ONTAP](#), a service that provides fully managed shared storage on AWS built on NetApp's popular ONTAP file system. Using this service, Phillips 66 enhances business agility, saves costs, and simplifies the management of its file storage.

Opportunity | Using Amazon FSx for NetApp ONTAP to Optimize Storage for Phillips 66

Founded in 1927, Phillips 66 is a multinational energy company that aims to provide energy and improve lives around the world. With a diverse portfolio of energy operations, including refining, midstream, chemicals, and marketing and specialties, Phillips 66 required technology that could support its dynamic business needs.

“Our leadership pushed us to be a cloud-first company and to migrate all our on-premises workloads to AWS,” says Taylor Savage, analyst at Phillips 66. “We needed to figure out a way to provide the same services in the cloud as we did on premises, and we also needed the migration to be as nondisruptive as possible.”

When the digital transformation was announced, there was no fully featured ONTAP filing system in any cloud. Phillips 66 ran all its file storage, in addition to its production workloads and DR environment, using NetApp ONTAP on premises. Everything from team file shares to drone images was hosted in this environment. With the introduction of Amazon FSx for NetApp ONTAP in September 2021, however, the company saw an opportunity for a simple, cost-effective migration without refactoring its apps.

The Phillips 66 migration to the cloud provided like-for-like ONTAP feature experience, from on-premises to AWS, which minimized the need for extensive retraining of its IT staff. The company could also scale more efficiently on AWS without significant operational overhead. Additionally, the managed upgrade and patching process provided by AWS would reduce the time and resources associated with these tasks.

“When Amazon FSx for NetApp ONTAP became available, it was a complete game changer for us,” says Savage. “We could use its features to address the challenges that we were facing on premises. We would no longer need to manage hardware or licensing, and we could take advantage of certain features to reduce costs.”

Solution | Migrating over 300 TB of Data from On-Premises to AWS in Less than 1 Month

To minimize disruptions to users, Phillips 66 took a strategic and phased approach to the migration. First, it focused on its DR environment, migrating over 300 TB of on-premises DR data to AWS in less than 1 month. Then, it began to migrate complete data center environments, including back-office applications and production workloads to [Amazon Elastic Compute Cloud](#) (Amazon EC2)—which offers secure and resizable compute capacity for any workload—and Amazon FSx for NetApp ONTAP. Phillips 66 relied on NetApp SnapMirror, a feature that replicates snapshot copies onto the cloud, to conduct these migrations in order to preserve its existing data



organization and security controls. By removing the need for code rewriting, rearchitecting, or refactoring, Phillips 66 simplified the migration of its vast and complex file [data architecture](#).

The migration was so seamless that it took one person—Savage—to handle the majority of the work. “We love how Amazon FSx for NetApp ONTAP is fully managed,” says Savage. “We don’t have to worry about managing all the underlying infrastructure, such as compute, storage, licensing, and networking. We’re already in the process of migrating additional NetApp workloads to AWS, including backup, DR, and data stores for our VMware Cloud on AWS deployments.”

With this setup, Phillips 66 maintains nearly continuous operations, even in the face of unforeseen events, by swiftly transitioning workloads to the cluster. The company also adopted elastic capacity-pool tiering to store its colder datasets. This feature of Amazon FSx for NetApp ONTAP offers a more cost-optimized way to store infrequently accessed data. Phillips 66 has cycled 90 percent of its data into the elastic capacity-pool tier, which has reduced its storage costs by 80 percent.

The cloud transformation has significantly enhanced Phillips 66’s business agility and the performance of its IT infrastructure, empowering the company to scale operations and respond to business needs promptly. For example, in 2023, Phillips 66 acquired a Denver-based company. And it had 2 months to exit the acquisition’s data center and migrate its applications and corporate file shares to the cloud.

In 1 week, Phillips 66 migrated over 132 TB of data to Amazon FSx for NetApp ONTAP, avoiding costly lease renewal. In another incident, an enterprise application for instrument monitoring went down at the company’s research center due to a patch disruption. Savage was tasked with quickly migrating the entire application’s production workload to Amazon FSx for NetApp ONTAP. In approximately 1 hour, he successfully migrated 330,000 files and resolved the issue.

Outcome | Bringing More Workloads to AWS to Improve Resilience and Performance

By adopting Amazon FSx for NetApp ONTAP, Phillips 66 has successfully modernized its file data architecture. The company is migrating its production workloads to Amazon FSx for NetApp ONTAP. Phillips 66 is also now moving its VMware clusters to [VMware Cloud on AWS](#), a managed service that combines compute, network, and storage capabilities in a fully supported, ready-to-run solution. Using Amazon FSx for NetApp ONTAP, it can size its VMware instances appropriately and scale storage as it grows—while optimizing for both performance and cost. The company can seamlessly use SnapMirror to replicate VMs and files from on-premises ONTAP volumes to Amazon FSx for NetApp ONTAP as supplemental datastores. This helps it to replicate entire refinery site clusters for DR capabilities in the cloud, serving as a failover in case of a disaster.

About Phillips 66

Phillips 66 is an energy manufacturing and logistics company with nearly 150 years of experience in the industry. The company has a vision to provide energy and improve lives around the world.

AWS Services Used



Amazon FSx for NetApp ONTAP

Amazon FSx for NetApp ONTAP provides fully managed shared storage in the AWS Cloud with the popular data access and management capabilities of ONTAP.

[Learn more »](#)

Amazon EC2

Amazon Elastic Compute Cloud (Amazon EC2) offers the broadest and deepest compute platform, with over 750 instances and choice of the latest processor, storage, networking, operating system, and purchase model to help you best match the needs of your workload.

[Learn more »](#)

VMware Cloud on AWS

AWS is VMware's preferred public cloud partner for all vSphere-based workloads. The VMware and AWS partnership delivers a faster, easier, and cost-effective path to the hybrid cloud while allowing customers to modernize applications enabling faster time-to-market and increased innovation

[Learn more »](#)

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